Neutron Source Facility Based on Compact Accelerators

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New technology in compact accelerators makes possible the development of high intensity non-reactor based neutron sources which are capable of performance levels comparable to or beyond that of reactors. The combination of new technology and the lengthy delays in the construction of conventional large neutron facilities offers a window of opportunity for the creation of university based facilities. Although reactors have long been the traditional source of neutrons, the production of neutrons by *spallation*, where a high energy proton beam from an particle accelerator is incident upon a high atomic number target has become a potential rival for reactors. The technical advantages of the use of spallation sources, particularly in time of flight techniques, have been widely discussed in the literature, however, their advantages as compared to reactors in non-technical areas because of the increasingly stringent regulation of reactors, their cost of operation and the difficulty of overcoming public political opposition to their construction may be equally important. Existing reactors are under increased pressure to close when their operating licenses come up for renewal, Further, new technology makes possible a new generation of spallation sources which are closer to the philosophy of smaller and more flexible research facilities. As will be shown, the spallation source offers the potential of producing neutrons at high rates with flexibility of use. This type of source also overcomes many of the problems associated with siting and operating reactors.

Neutrons have been used as powerful tools for many investigations in science and engineering. There are currently proposals for several large neutron facilities in the US. which would provide world-class neutron sources to compete with those existing in Europe and Japan. Unfortunately, these facilities also carry with them large, world-class budgets and long construction times. For example, the Advanced Neutron Source, to be built at Oak Ridge has an estimated cost of $3 billion and a completion time estimated to be the year 2000 or beyond. In recent years, several large projects such as the Superconducting Supercollider (SSC) have been halted or, as with the Space Station, severely curtailed in scope and almost halted. Large single projects are becoming increasingly harder to fund over multi-year time periods and end users are finding it more difficult to sustain their research programs while waiting for the project to be completed. In contrast to the massive projects, new technology has made a different design philosophy possible so that increasing emphasis is placed on development of multiple, relatively small, specialized systems rather than a single massive system. Examples include the trend toward distributed computation and use of workstations and away from mainframes, NASA's moving away from large multi-purpose satellites toward smaller, lower cost satellites, and finally, the work at MIT on small, autonomous robots as new ways for exploration of the ocean and of space.

Aside from the general problems inherent in large high-visibility projects, large reactors have their own set of problems. They are increasingly more difficult to site and to operate due to environmental and political concerns. The use of highly enriched fuels in almost all of the new reactors has in itself become a subject of concern due to issues of weapons proliferation. Current designs are pushing the limits of technology with respect to reactor core power density, and since core power density sets the limit on external beam flux this limit is also being reached. Further, it is difficult to operate such reactors in pulsed mode. In the next few sections some of the technical aspects of these new small spallation sources will be discussed, but a more careful feasibility study should be initiated quickly to determine the practicability of this approach.

Neutron Source Facility Based on Compact Accelerators: Technical Considerations

Introduction

Recent technical developments, primarily in *compact high current ion accelerators* originally developed for the SDI program, open the way for a new generation of spallation sources which are considerably more compact and less expensive than the current generation of spallation sources. As shown in Table 1, most of these current sources are based on large, high energy accelerators, typically 500 MeV or more which produce usable neutrons at rates of 10^{16} or more, comparable to *usable* rates from reactors. In most cases, these machines are based on older accelerators which were retired from work in nuclear physics and are not necessarily optimized for use as spallation sources. In this note, a different approach to spallation sources based on low energy, high current accelerators will be discussed and some estimates made of the production rates for neutrons.

Facility	Characteristics	Average fast-neutron production rate (n/s)	Source pulse width (μs)	Pulse repetition frequency (Hz)	Status
		Spallation sources			
ZING-P (Argonne, USA)	200-MeV protons, Pb target	5.0×10^{11}	0.2	10	1974–1975 (closed)
ZING-P′ (Argonne, USA)	500-MeV protons, W, U targets	$2.4,5 \times 10^{14}$	0.2	30	1977–1980 (closed)
IPNS (Argonne, USA)	12-μA, 500-MeV protons, U target	1.5×10^{15}	0.1	30	1981– (operating)
ASPUN (Argonne, USA)	3.8-mA, 1.5-GeV protons, U target	1.5×10^{18}	0.14	100 (multiplexed)	1990s (proposed)
SNS (Rutherford, UK)	800-MeV protons, U target	4.0×10^{16}	0.27	50	1984– (operating)
LAMPF-WNR (Los Alamos, USA)	800-MeV protons, W target	$(1\text{–}2) \times 10^{15}$	~ 5	120	1977– (converted)
WNR/PSR (Los Alamos, USA)	100-μA, 800-MeV protons, W target	1.2×10^{16}	0.27	12	1985– (operating)
KENS (Tsukuba, Japan)	500-MeV protons, U target	3.0×10^{14}	0.07	15	1980– (operating)
KENS II (Tsukuba, Japan)	0.5-mA, 0.8-GeV protons, U target	1.0×10^{17}	0.5	50	1990s (proposed)
SNQ (KFA Jülich, W. Germany)	5-mA, 1.1-GeV protons, U target	1.5×10^{18}	250	100	1990s (proposed)
SINQ (SIN Villigen, Switzerland)	1-mA, 0.6-GeV protons, Pb–Bi target	7.0×10^{16}	Continuous		1988 (proposed)
TTNF (TRIUMF, Vancouver, BC, Canada)	150-μA, 0.5-GeV protons, Pb target	8.0×10^{15}	Continuous		1978 (operating)

Table 1. Current and proposed spallation sources (1986)

Spallation sources

The measured yield of neutrons in neutrons/incident proton, Y, is given by [1]:

$$Y(E) = 0.1\,(E - .12)(A + 20)$$

and, for U^{238}, is given by:

$$Y(E) = 50\,(E - .12)$$

where E is the incident proton energy in GeV and A the atomic weight of the target. Figure 1 show this for thick targets with various bombarding particles and Figure 2 shows results for protons on several targets at high energies.

1. J. M. Carpenter and W.B. Yellon, Meth. Exp. Phys. v 23. pp 99 - 196 (1986)

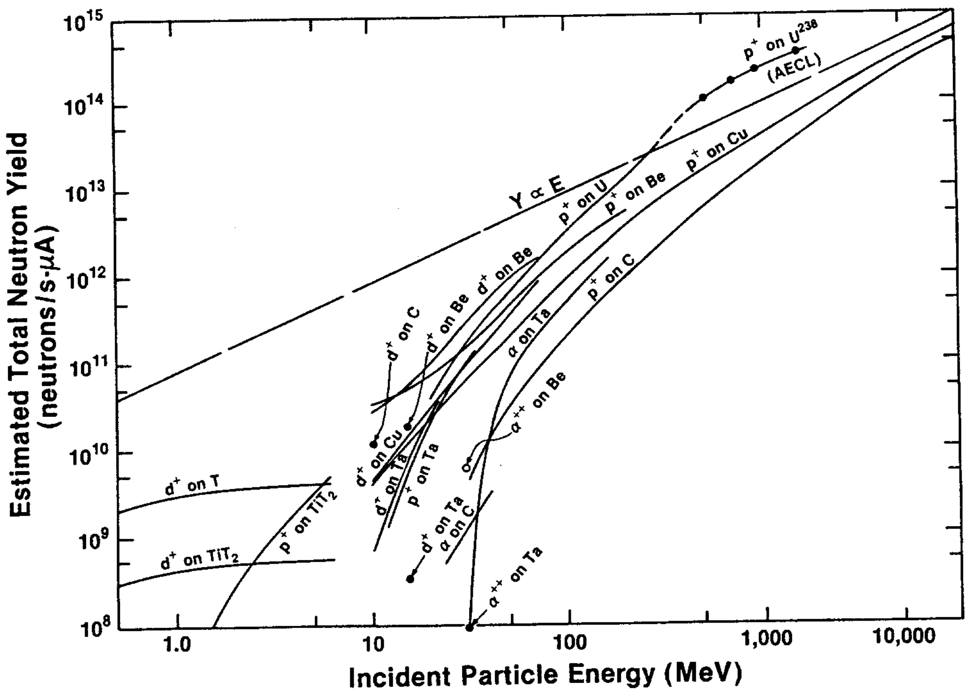

Figure 1. Thick target yields

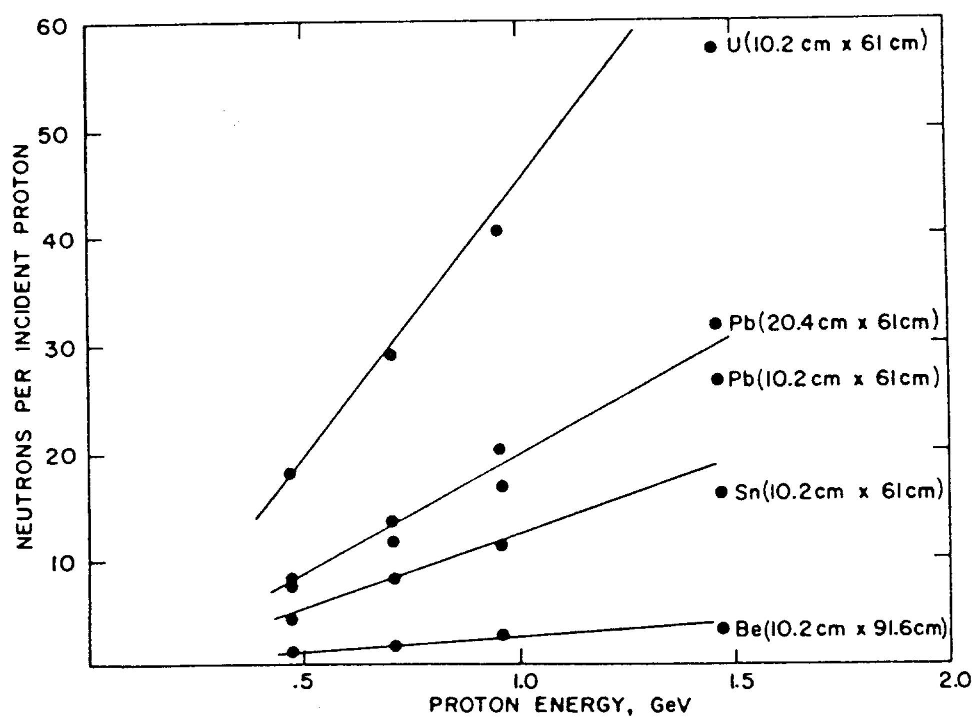

Figure 2. Neutron yields for incident protons

The total neutron production will just be the yield multiplied by the incident proton intensity. The production of neutrons as a function of incident power vary as:

$$N \propto (1 - .12/E)$$

and thus is only slowly varying with incident energy once we are much above 0.12 GeV. The major considerations are the ability of the accelerator to deliver power to the target and also the ability of the target to dissipate the heat.

A possible figure of merit for the optimum energy and current may be made by calculating the neutrons/Watt on target. The incident number of protons is given by:

$$N_p = i/q$$

$$= 6 \times 10^{15} \text{ protons/mA}$$

which gives a power of:

$$P(W) = 10^6 E(\text{GeV}) \; i \,(\text{mA})$$

the number of neutrons/sec is therefore (for U^{238}):

$$N_n = Y N_p$$

$$= 50 (E - .12) i/q$$

$$= 3 \times 10^{17}(E - .12) i$$

and the ratio of neutrons/W is:

$$R = 3 \times 10^{11}(E - .12)/E \text{ neutrons/W}$$

The energy distribution of the neutrons is very similar to that from fission [2]:

$$N(E) = E^{1/2} e^{-E/T}$$

where T is typically 1.3 MeV. This gives the well known result that the peak of the distribution is at 1/2 T, 650 keV, and the average energy is 3/2 T or about 2 MeV At incident proton energies of several hundred MeV only about 2% of the neutrons produced are high energy and are peaked forward with energies up to the incident beam energy. It should be noted that this high energy neutron tail will however dominate the shielding requirements for the system and will be an important issue in overall system design. Aside from this, the target looks like a spatially small fission source.

None of the above is particularly new, however the development of high current, i.e. mA beams rather than μA beams typical of earlier accelerators means that comparable neutron source strengths

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[2]. ibid.

can be obtained with lower energy accelerators and hence considerably smaller and less costly installations, limited primarily by target heat limitations.

Two examples of this new technology are the 140 MeV, 2.5 mA, RFQ proposed by AccSys Technology [3] and a similar cyclotron proposed by IBA [4] Based on these numbers we obtain a value for Y of 1.0 neutron/proton for a U^{238} target. The number of protons per mA is approximately 6 x 10^{15}, so these sources will produce approximately 1.5 x 10^{16} n/s, comparable to the *usable* neutrons from reactors. [As an example, a 10 MW reactor produces a *total* of approximately 10^{18}/n/s, but only a limited fraction of these, perhaps 5% are actually available since the reactivity of the core must also be maintained, reducing the *usable* number of neutrons to 5 x 10^{16} n/s.] The physical size of an RFQ is approximately 30 m long while the cyclotron is approximately 4 m in diameter. Note that in both cases, the beam may be transported from outside the target area and thus will result in a very compact volume to be shielded.

In practice, a trade-off must be made between beam current and total energy as it may often be easier to increase energy than to increase current. Figure 3 shows the current and beam power for a constant neutron output, where power and current are normalized to 1.0 for 0.14 GeV beam energy.

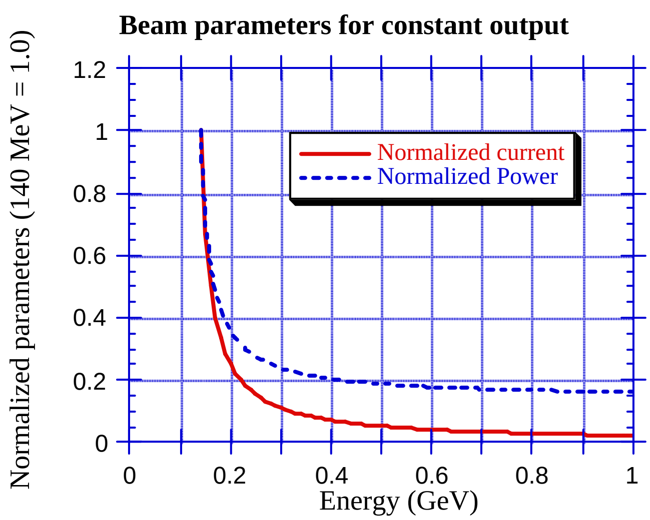

Figure 3. Normalized beam parameters for constant neutron output

For some applications, e.g. boron neutron capture therapy (BNCT), accelerator based systems producing 10^{12} n/sec have been proposed [5] using the reaction (p,Li^7) to produce neutrons using a 2.5 MeV, 4 mA beam. There are however a considerable number of technical problems which must be

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3. AccSys Technology, Pleasanton, CA. It should be noted that only the first few MeV of such an accelerator is based on the RFQ technology and that subsequent stages are more conventional LINAC designs, however the term RFQ will be used to denote such a mixed system.

4 Y. Jongen, T. Delvigne and P. Cohilis, "Multi-milliampere Compact Cyclotrons used as Neutron Sources", 4th International Conference on Applications of Nuclear Techniques", Crete, June 1994 (in press)

5. Science Research Laboratory, Somerville, MA

overcome before these currents are obtained at this energy with this target and these numbers remain as goals. Interestingly enough, the proton therapy facility under construction at MGH could also be used for BNCT. This system is based on an IBA manufactured 235 MeV cyclotron with an external beam current of 300 nA. The limitation is established to ensure patient safety and is limited at the ion source. There is an experimental area where the external beam may be brought out without the magnet gantries for patient treatment found at the three treatment areas. Based on these numbers and on a U^{238} target, we find that Y = 5.75 n/p and neutron production rate is 10^{13} n/sec. The power dissipated in the target is only 70 W and little cooling would be required, thereby permitting a very compact installation with a minimum patient-source distance. Although the energy spectrum is not the same as that from the (p,Li) reaction, it appears that such a system would be quite competitive with the proposed SRL system.

To fully compare these systems with reactors will of course depend on the final applications. The pulsed nature of RFQ accelerators in which the peak flux may be as much as 50 x the average leads to important advantages for these devices as neutron sources: As an example, for neutron diffraction work, a spallation source based on a pulsed system, such as an RFQ, opens up in a natural way the use of time of flight (TOF) information for neutron energy determination in Laue diffraction. Figure 4 illustrates the differences between using a reactor source and a pulsed spallation source for neutron diffraction. In the case of energy monochromatization using a crystal, only a narrow range of wavelengths is selected at a time whereas the time of flight technique selects many wavelengths simultaneously, increasing the effective data taking rate (although the number of neutrons/s at any *particular* wavelength is lower).

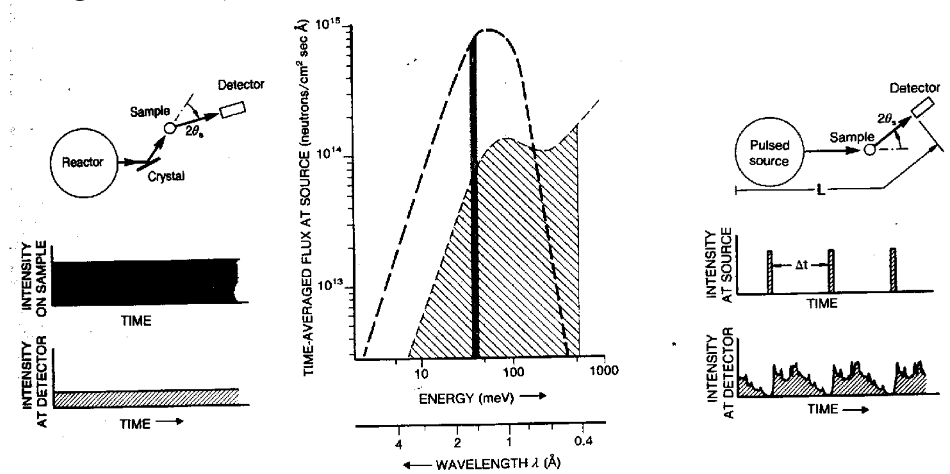

Figure 4. Comparison of pulsed versus steady source

Mezei has recently pointed out that in TOF diffractometry, for ideally designed instruments requiring neutron monochromatization, the time averaged flux of a pulsed machine equals its peak flux, and thus the effective data gathering power of pulsed sources is equivalent to much larger steady state sources. [6] This means that the equivalent continuous reactor source strength required is given by multiplying by 1/(duty factor). Since RFQ sources normally operate with duty factors of 2 - 4%, the multiplication is 25 - 50 compared to the pulsed system. This is shown in Figure 5 for various spallation and reactor sources.

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6. F. Mezei, "On the Comparison of continuous and pulsed sources". Neutron News, v 5, No. 3, p 2, (1994)

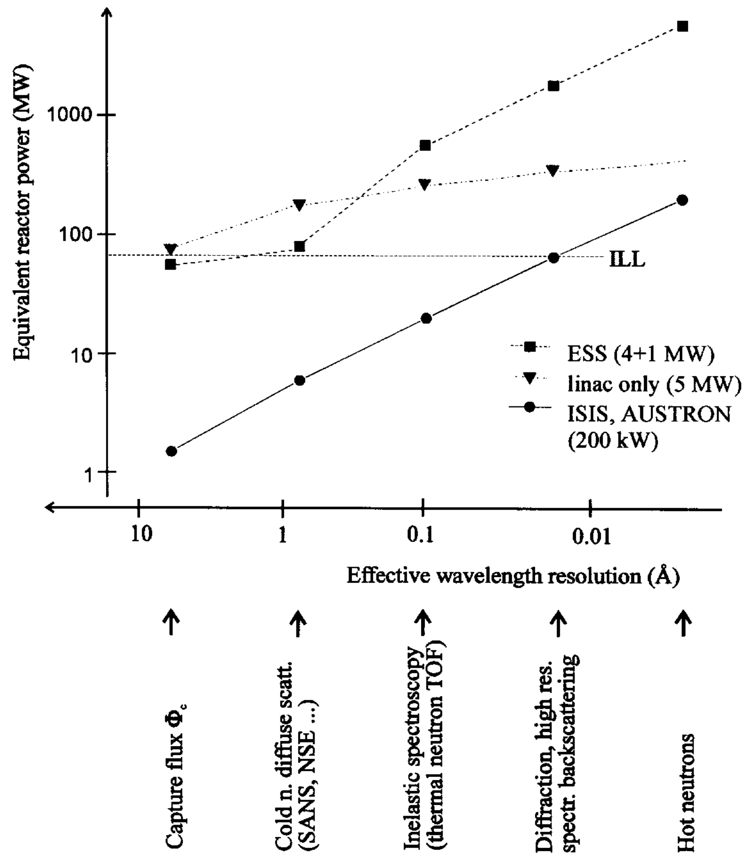

Figure 5. Equivalent reactor powers for spallation facilities

Another interesting possibility, especially for RFQ accelerators with high current capability, is time multiplexing of the beam using switching magnets as shown in Figure 6. Further, by shutting off sections of the accelerator or deflecting the beam out between LINAC sections, it is possible to also change energy on a pulse to pulse basis. Thus various experiments can be run with different targeting and with different energies. It might be possible to use a low energy pulse at a low effective rep rate to simultaneously produce isotopes for PET (positron emission tomography) or other nuclear medicine applications with little sacrifice of total neutron flux. Production rates of various isotopes as a function of proton energy are shown in Figure 7.

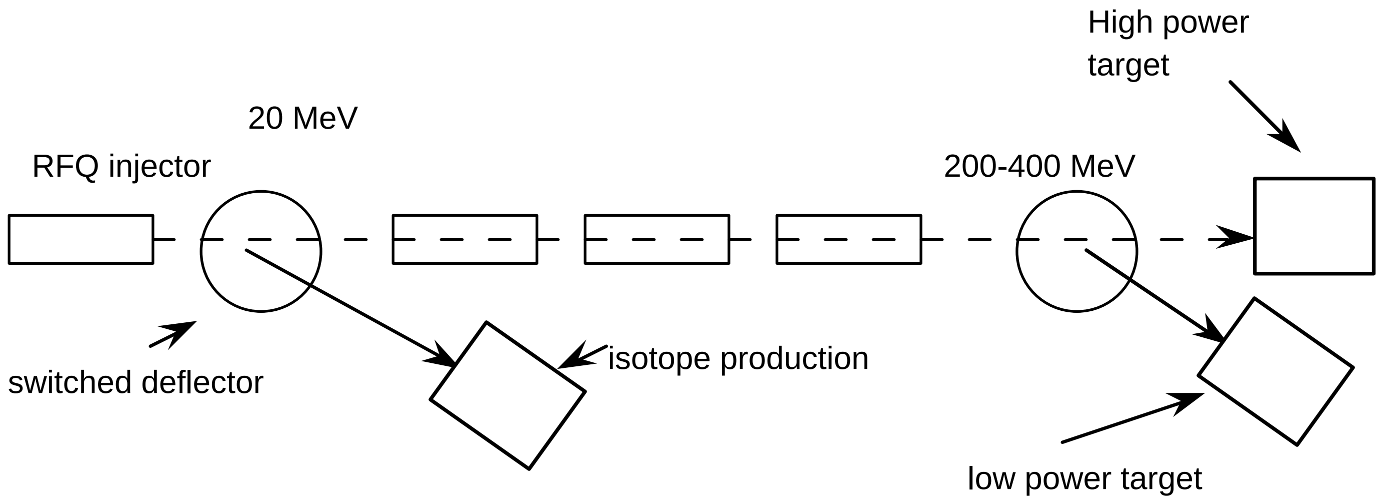

Figure 6. Example of spallation system

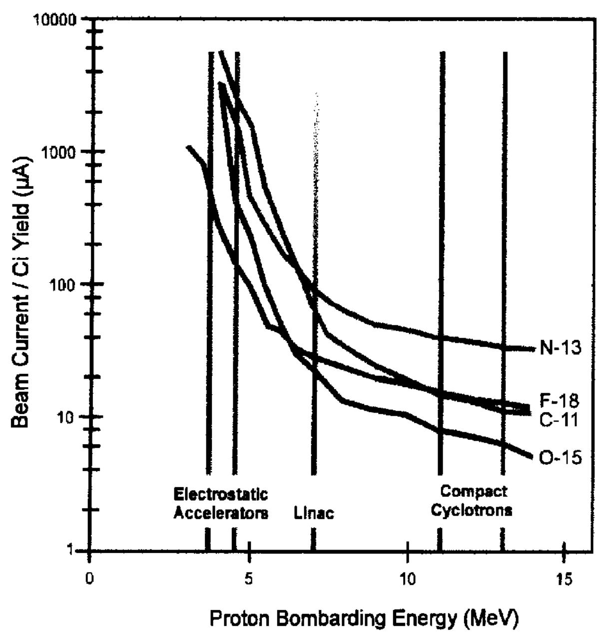

Figure 7. Isotope production rates

The central design difficulties will be in the target design and the shielding of the high energy neutrons. These are areas in which considerable expertise exists at MIT.

To cite one example, the heat load on the target could be as much as 1 MW, concentrated in a relatively small volume. The dissipation of this heat requires careful investigation. For example, at the expense of decreased neutron output, a liquid metal target composed of lead/bismuth mixtures is a potential approach which has been investigated at the Paul Scherrer Institute (PSI) for a 1 MW liquid metal target. [7] As another possibility , the design of divertors in plasma fusion systems have produced extensive developments in methods for removing large heat loads, as large as 100 MW/m^2

[7]. Y. Takeda and W.E. Fischer, "Thermohydraulic Behavior of the Liquid-Metal Natural-Circulation Target of the Spallation Neutron Souurce at Paul Scherrer Institute", Nucl. Sci. and Eng. 113, No 4, p 287 (1993)

using liquid cooled systems and also have resulted in new developments using helium cooled high heat load systems.

Reactor designers have had many years of shielding design experience which would be required as part of the overall system design and similar breadth of experience in neutronics for producing useful beams and neutron transport systems. In this regard, it is possible to consider a situation where an existing reactor core is replaced by a spallation target, making it possible to use the extensive shielding and beam port design in the existing facility. If a sub-critical assembly were employed, the neutron yields calculated earlier could be increased by a factor of ten or more.

For completeness, Table 2 is included which shows projected yields from a variety of commercially available accelerators. With the exception of the IBA cyclotron, they are generally considerably lower in neutron output than the spallation sources which have been discussed. They do however point toward intermediate stages in a spallation source design where low energy machines can be used as stand alone neutron sources and then as injectors into higher energy machines.

Generator or Accelerator Model	Incident particle	Target Element	Incident particle energy [MeV]	Beam current [μA]	Total Yield [neutron . s^{-1}]	Energy Deposition (MeV/n)	Mean Neutron Energy (MeV)	Thermal Peak Flux [neutron . cm^{-2} . s^{-1}]	"S" (Thermal Peak Flux / Watt)
TN 46 Tube (Sodern)	d	Tritium	0.225	10,000	4 10^{11}	35,000	14	1.2 10^{9}	5.3 10^{5}
PL-4 Linac (Accsys)	p	Be	3.9	1,000	1.2 10^{12}	20,300	0.5	2.6 10^{10}	6.6 10^{6}
Cycl. 3D (IBA)	d	Be	3.8	2,000	5.7 10^{12}	8,400	1.95	6.4 10^{10}	8.4 10^{6}
Cycl. 3D-7P	p	Be	7	2,000	2 10^{13}	4,375	1.25	3.3 10^{11}	2.4 10^{7}
Cycl.18/9	p	Be	18	100	1. 10^{13}	1,070	4	6.5 10^{10}	3.6 10^{7}
Cycl. 18+	p	Be	18	2,000	2.1 10^{14}	1,070	4	fast n only	fast n only
Cycl. 30 (IBA)	p	Be	30	400	1.3 10^{14}	580	7	6.11 10^{11}	5.1 10^{7}
Cycl. 30	p	Be	30	800	2.6 10^{14}	580	7	1.22 10^{12}	5.1 10^{7}
Cycl. 140 *(IBA)*	p fission	Pb & ^{235}U	140 slow n	1,000	6 10^{15} (> 2 10^{16} for k=5)	145 ~ 30	3 (slow n)	5 10^{13} 2.5 10^{14}	3.6 10^{8} 1.8 10^{9}
Rhodotron (IBA)	e-	^{238}U	15	10,000	2. 10^{14}	4,700	2	2.4 10^{12}	1.6 10^{7}

Table 2. Accelerator based neutron sources.

The design and fabrication of a spallation source clearly involves major costs, including initial capital cost as well as operating costs. Some studies have shown that even very large spallation facilities have considerably lower personnel requirements and therefore lower overall operating costs. Part of the capital cost of such a facility could be reduced by locating the target in the core of the current reactor and thus using both the building as well as the reactor itself as parts of the spallation source. Further, the beam could be switched to other target areas for such applications as isotope production, radiation therapy (both protons and neutrons), and production of fast neutrons.